



Advanced Digital Signal Processing

Lecture 9

Polyphase Filters

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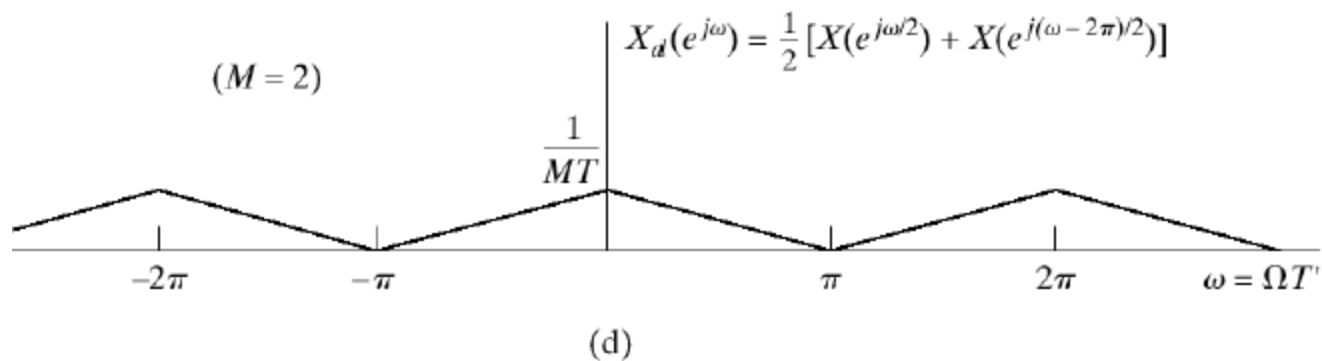
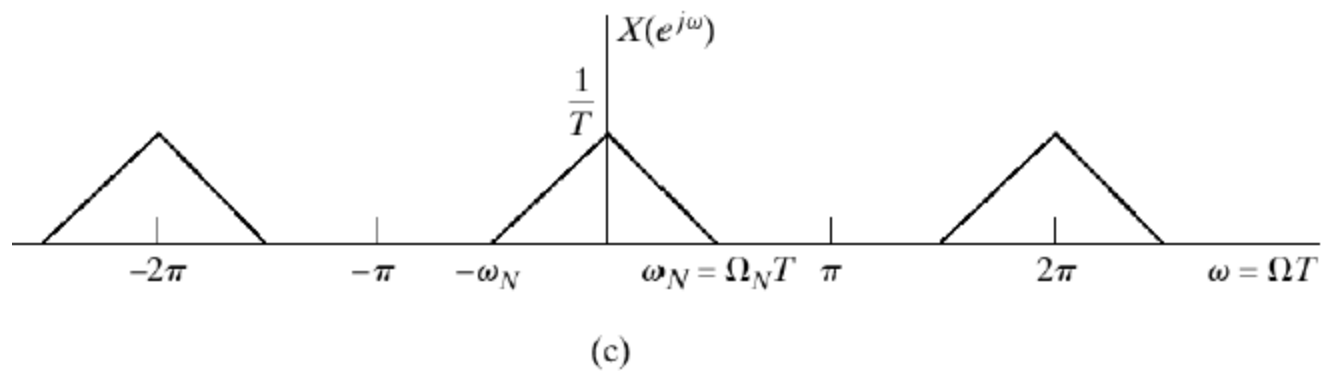
Multirate Signal Processing

- The two primary operations in multirate processing are:
 - Decimation (Down sampling)
 - Interpolation (Up sampling)

Changing the sampling Rate

- It is often advantageous to digitally alter the sampling rate during processing, after the initial A/D conversion has been finished:
 - High sampling rates (perhaps many times the Nyquist rates) can be used to remove unwanted frequency components
 - Low sampling rates (near the Nyquist rate) are best for computation speed and memory requirements

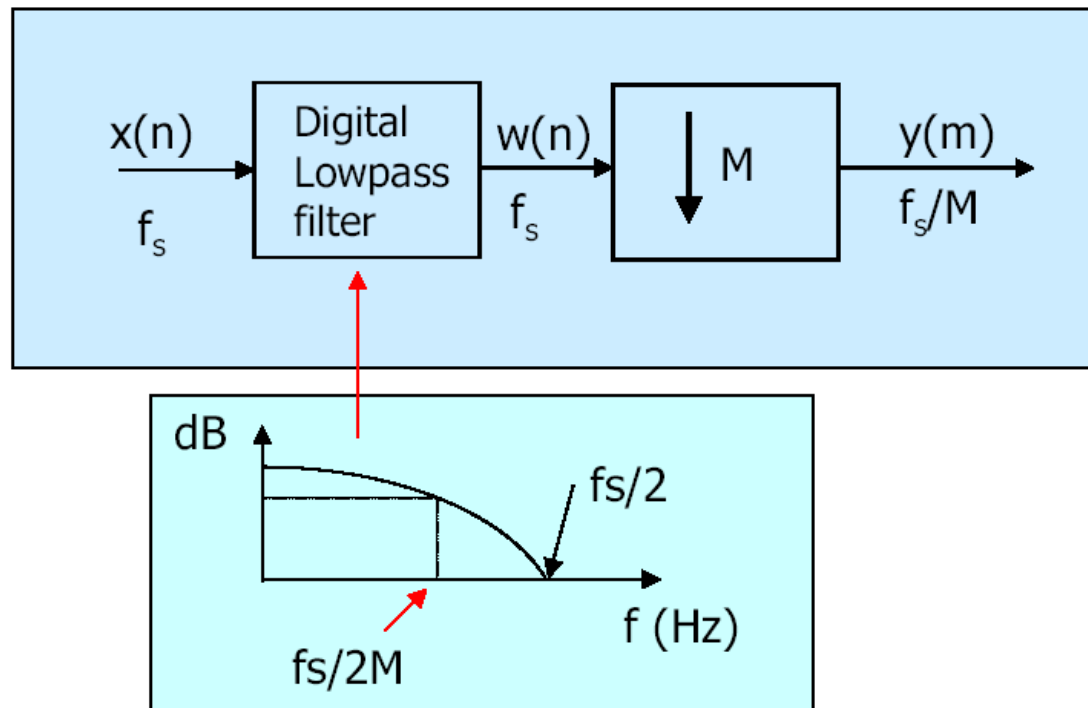
Down sampling



Decimation Filter

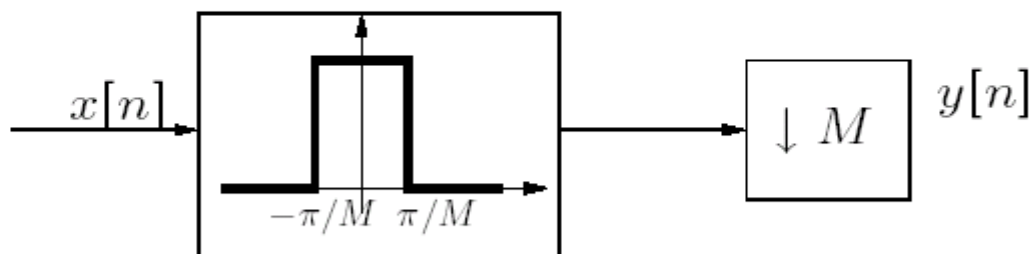
The diagram below shows a block representations of a times M decimator.

The signal $x(n)$ is first passed through a lowpass filter that attenuates the band from $f_s/2$ to $f_s/2M$ to prevent aliasing.



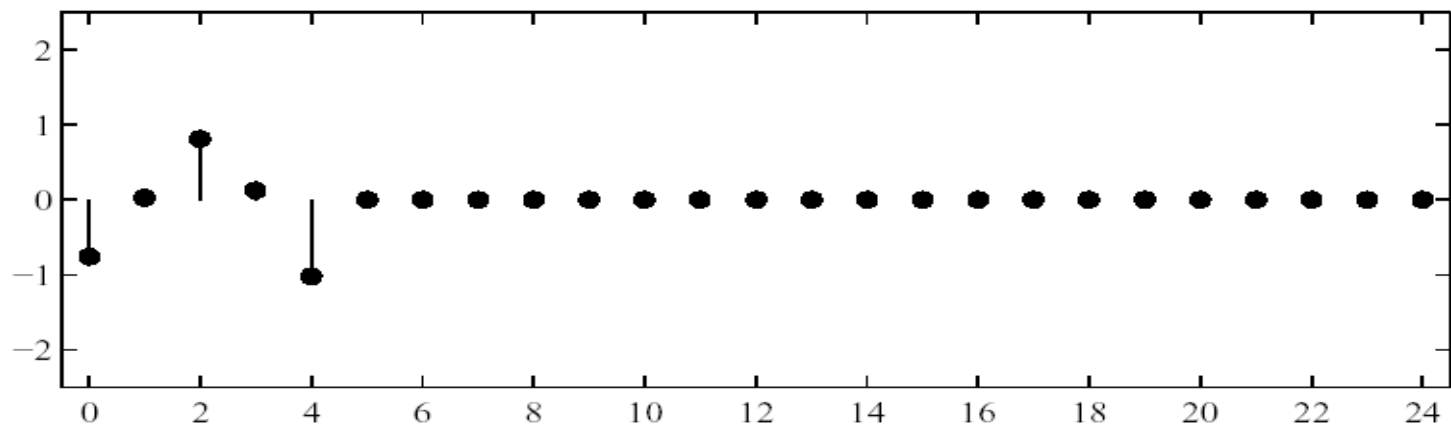
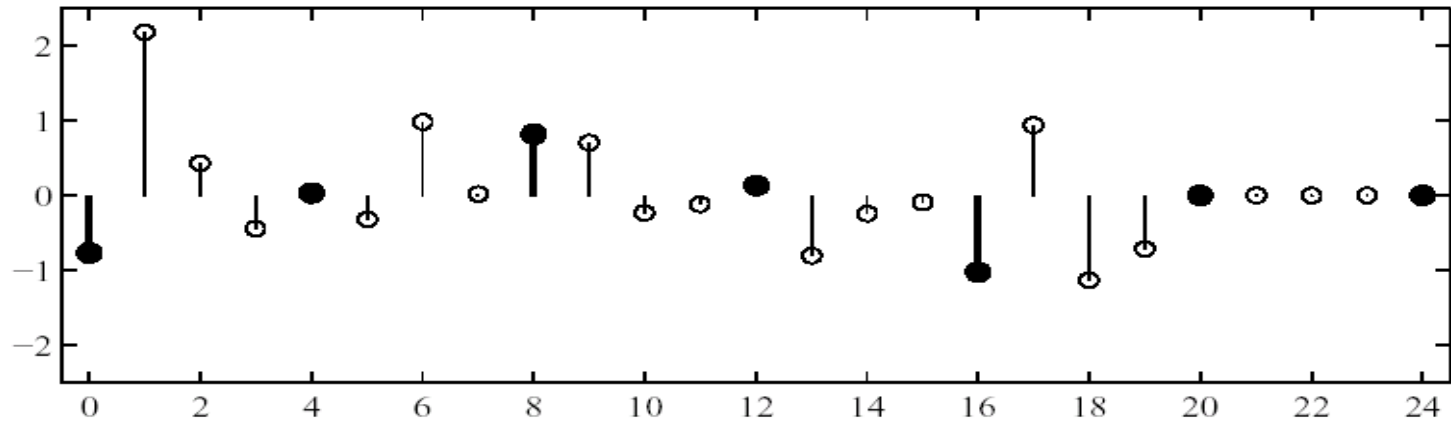
Aliasing and pre-filtering

- No aliasing if $X(e^{j\omega}) = 0$ for $\frac{\pi}{M} \leq \omega \leq \pi$
- If not satisfied: pre-filtering is needed:



Note: in this case, $y[n] \neq$ samples of $x(t)$!!

Downsampling in time-domain

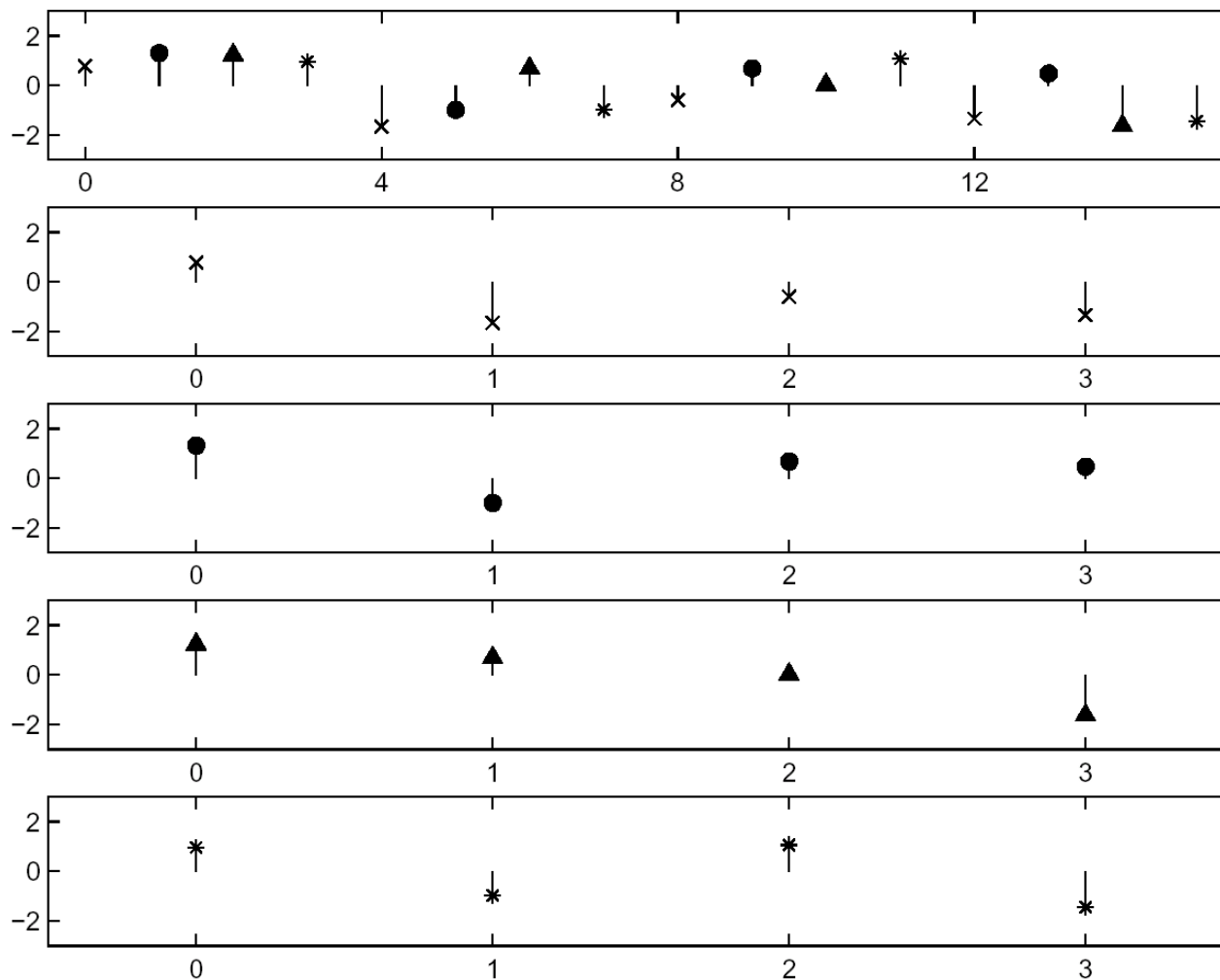


deci



Polyphase decomposition for decimation

$$h[n] \longrightarrow e_k[n] = h[nM + k]$$



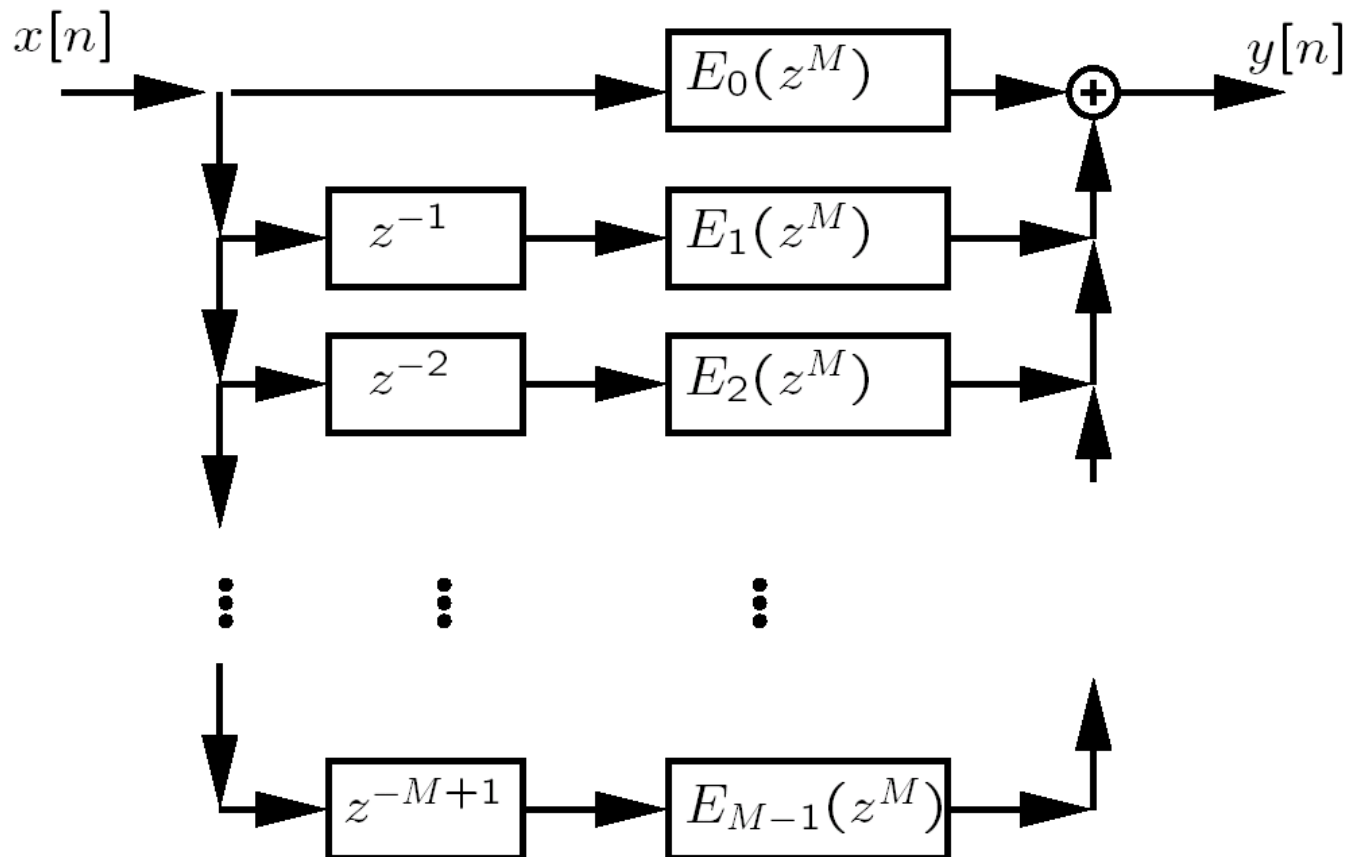
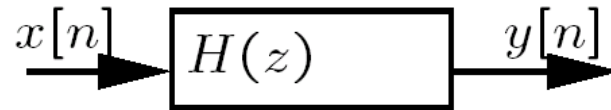
Polyphase decomposition

$$h[n] \longrightarrow e_k[n] = h[nM + k]$$

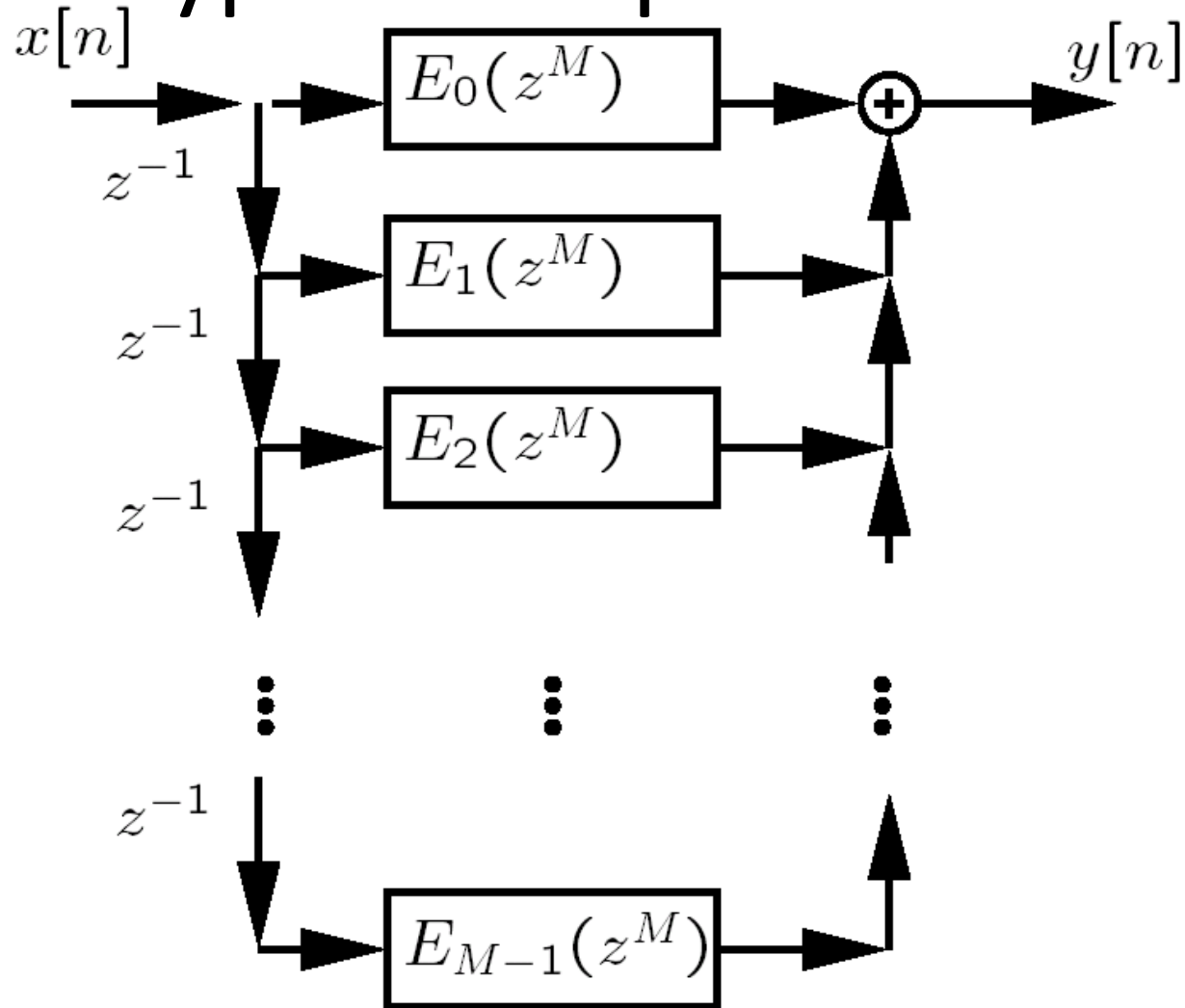
$$E_k(z) = \sum_{n=-\infty}^{+\infty} h[nM + k] z^{-n}$$

$$\begin{aligned} H(z) &= \sum_{m=-\infty}^{+\infty} h[m] z^{-m} = \sum_{k=0}^{M-1} \sum_{n=-\infty}^{+\infty} h[nM + k] z^{-nM-k} \\ &= \sum_{k=0}^{M-1} z^{-k} E_k(z^M) \end{aligned}$$

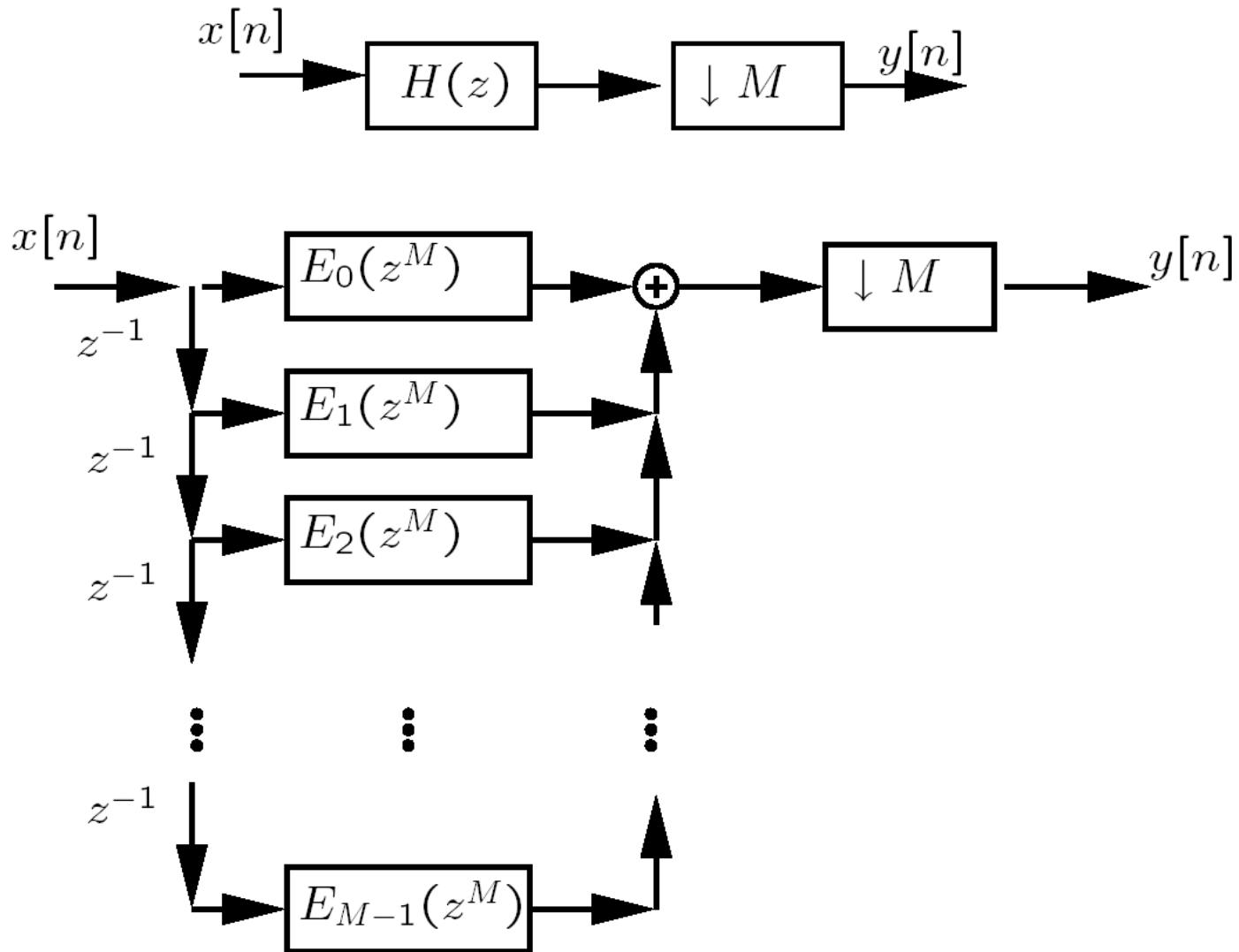
Polyphase representation



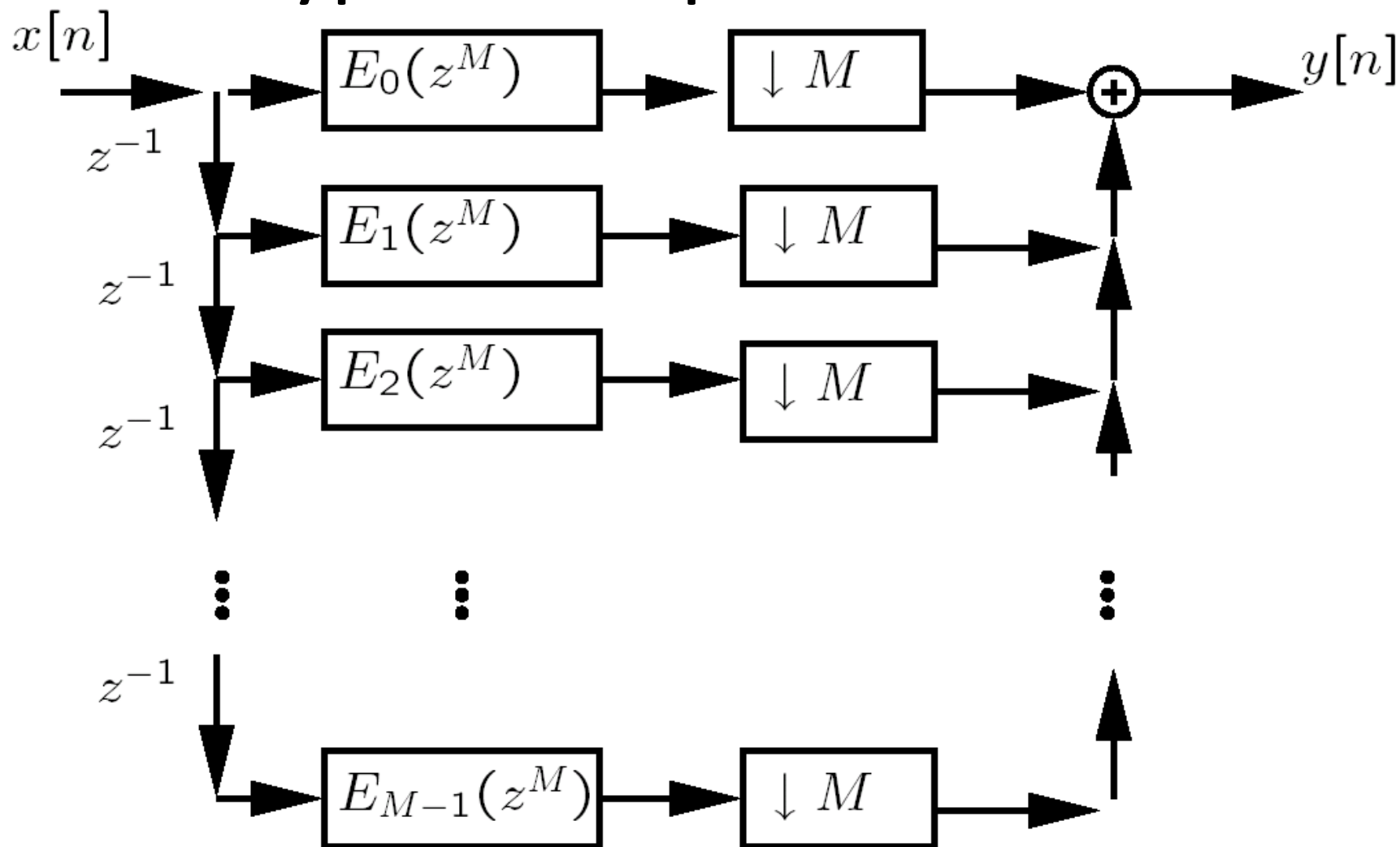
Polyphase Representation



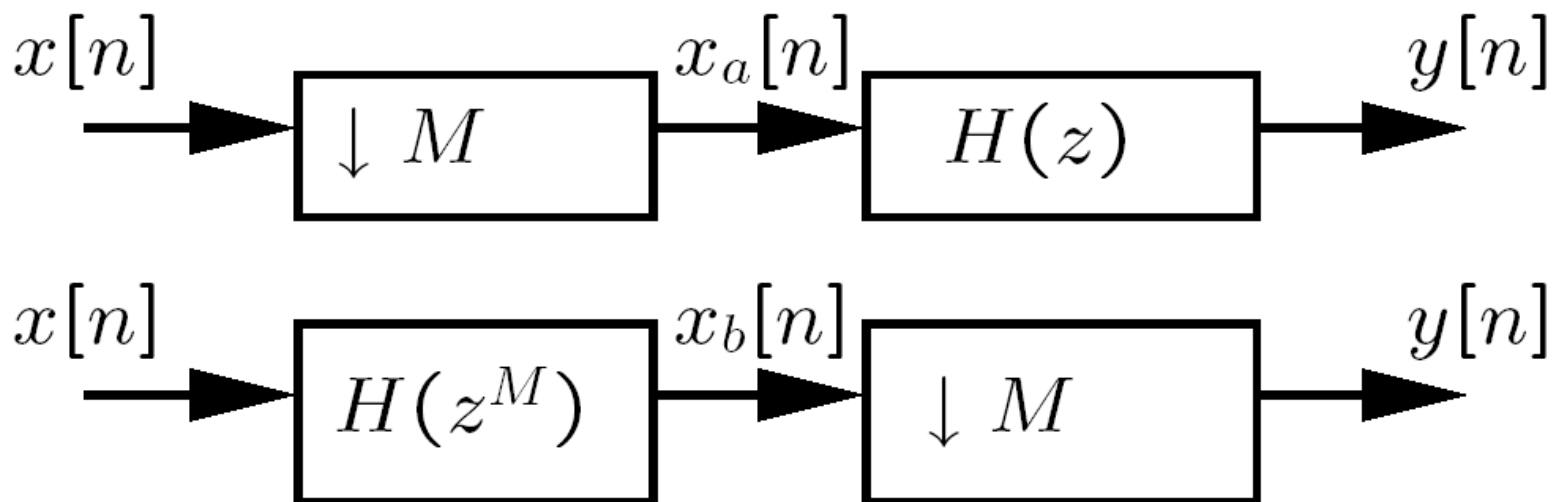
Polyphase Representation



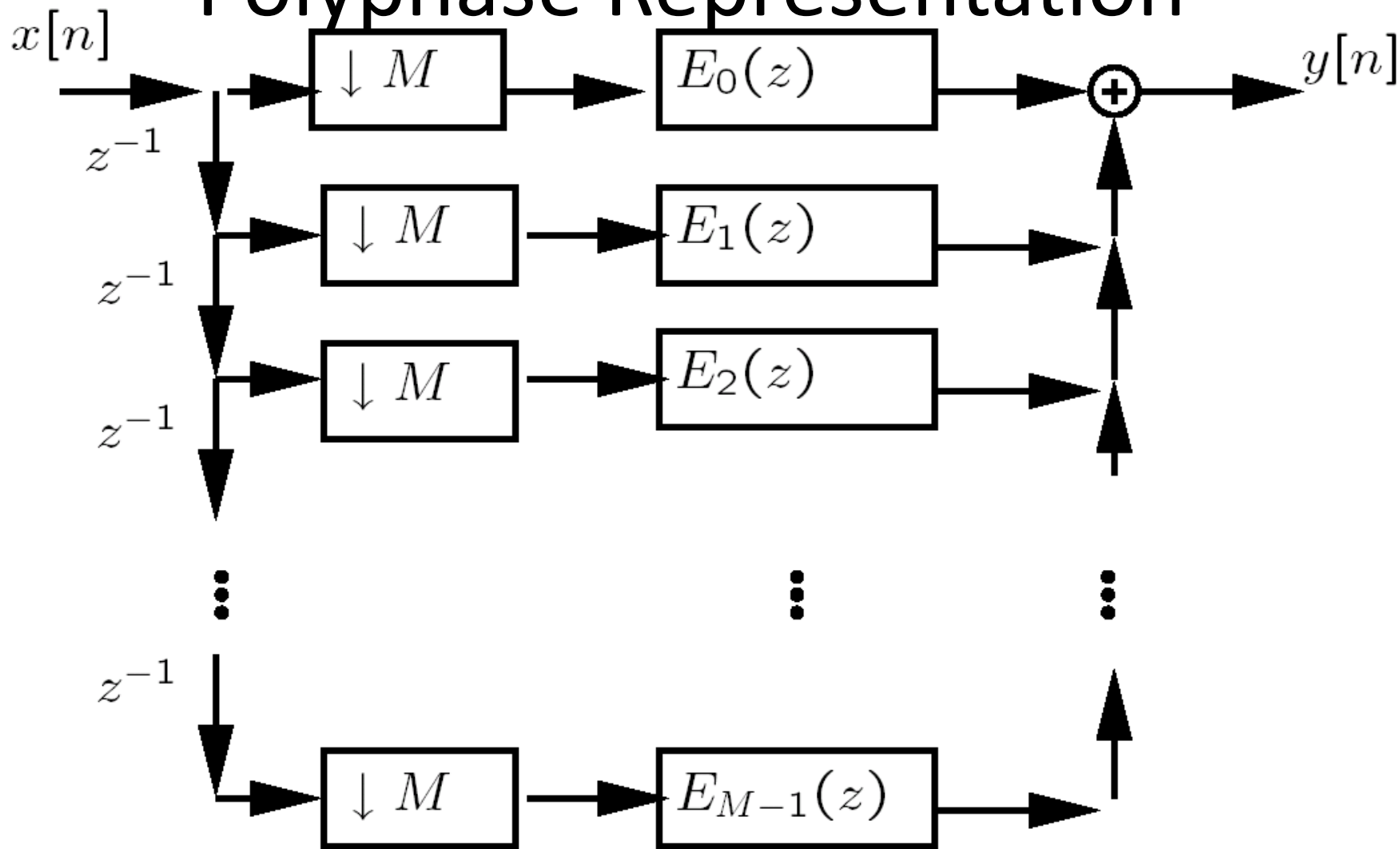
Polyphase Representation



Noble Identities



Polyphase Representation



Decimation Filtering: IIR Vs. FIR

Should we use IIR or FIR for the lowpass filtering required?

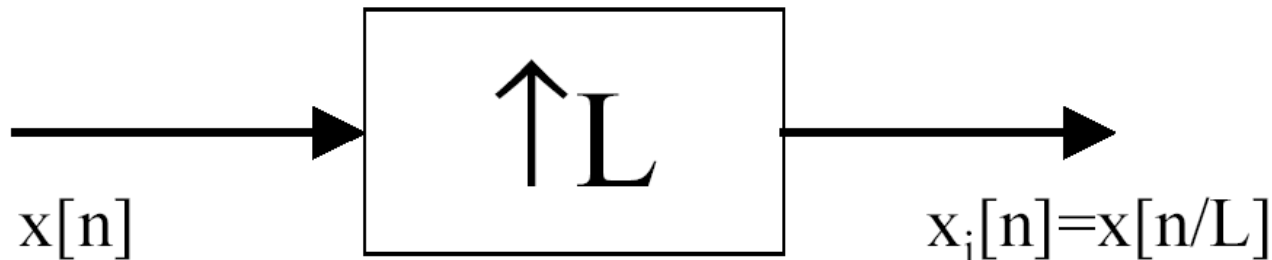
Using an IIR filter in this case has an obvious shortcoming. We cannot take advantage of the fact that we only have to compute every N th output, since previous outputs are required to compute the M th output. Thus no saving is realised.

On the other hand, using an FIR filter, in this case implies that we can do our computations at the rate of f_s/M . Thus, using an FIR filter in the decimation process will lead to a significantly lower computation rate.

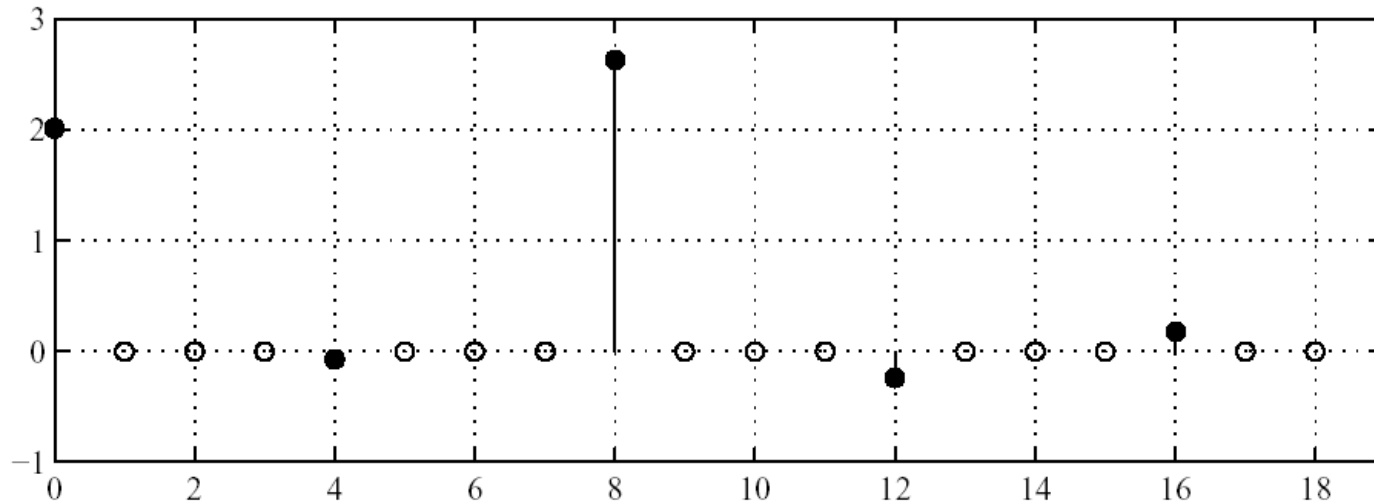
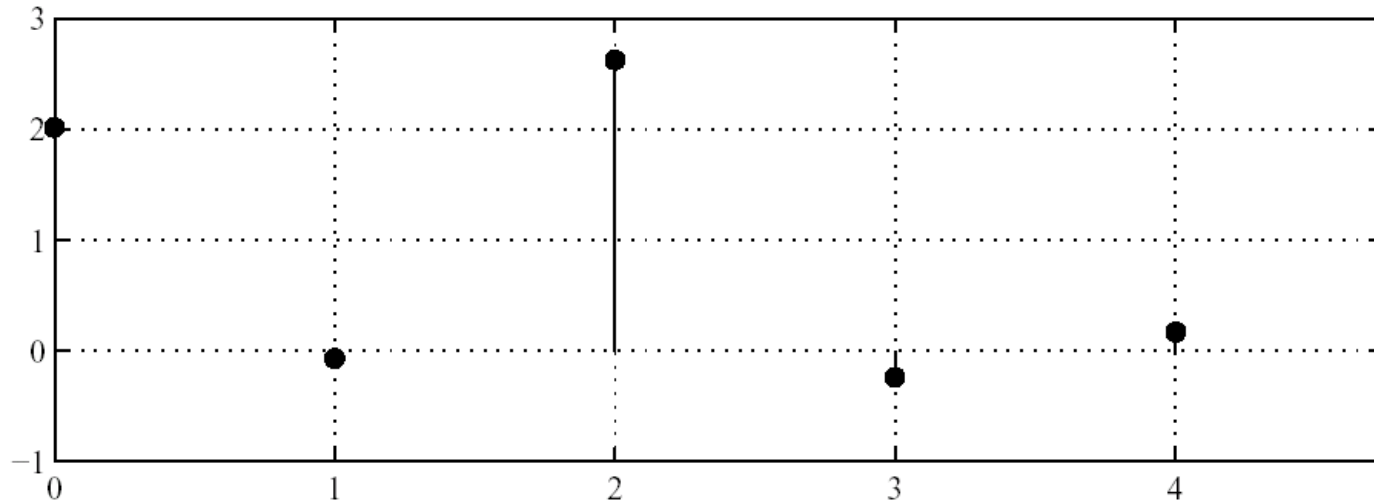
Another advantage of using an FIR filter is the fact that we can easily design linear phase filters and this is desirable in many applications.

Upsampling

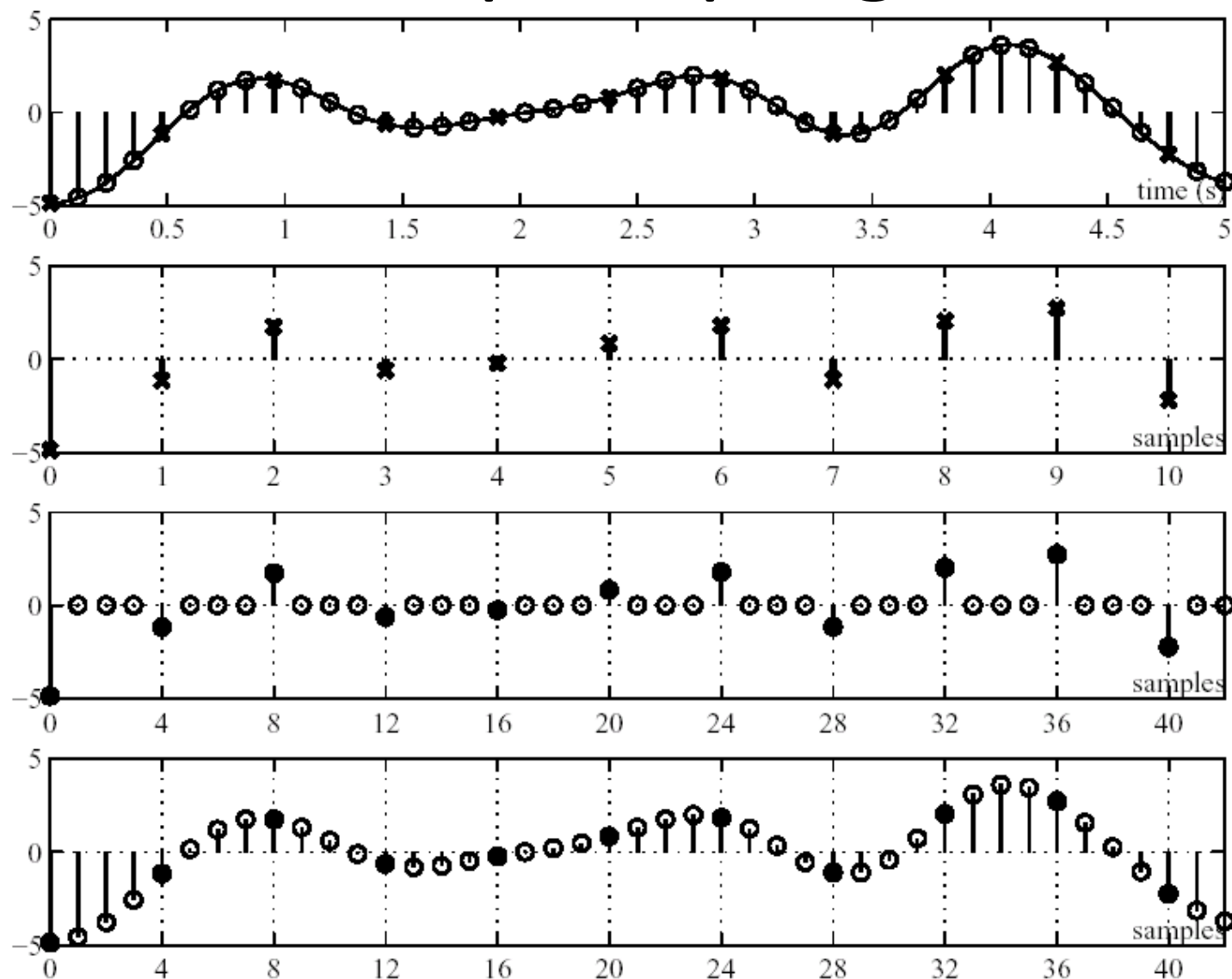
- ***Upsampling*** by an integer factor L (may also see I used) is essentially the process of inserting $(L-1)$ new values between every sample. This spreads the samples out and is thus sometimes called an *expander*.



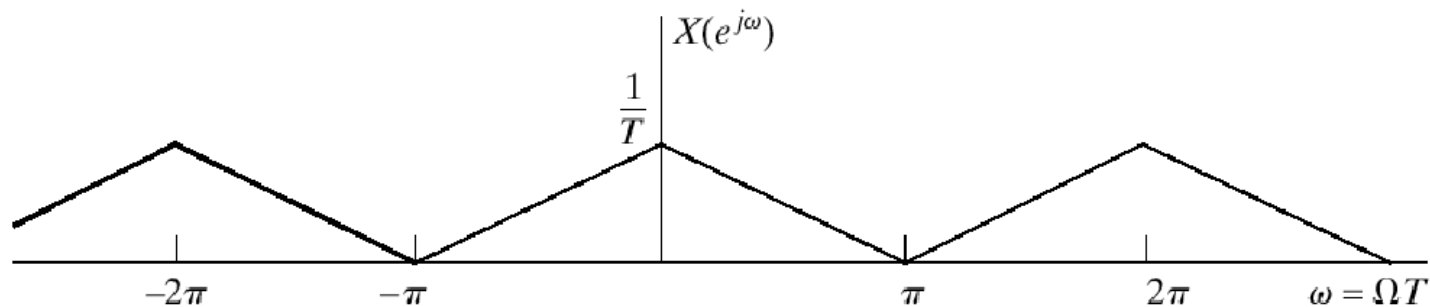
Upsampling in Time



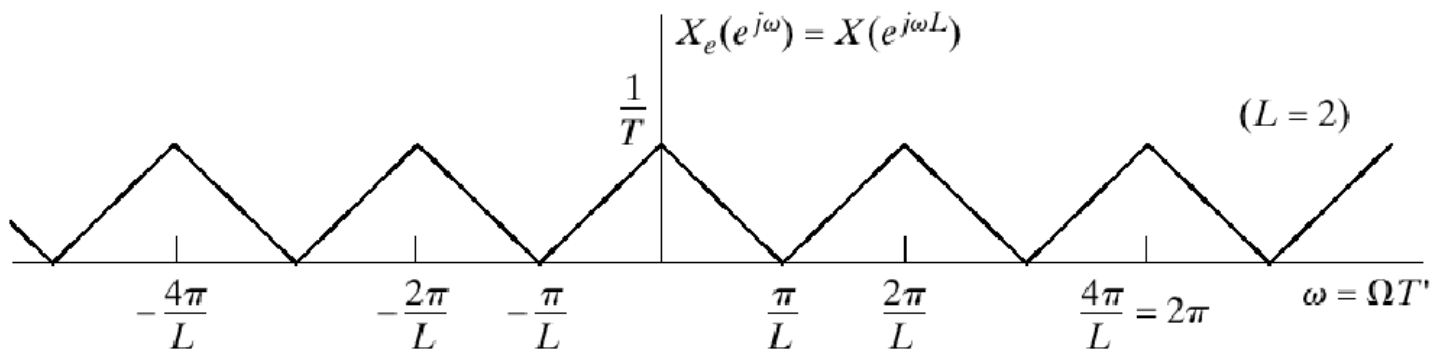
Upsampling



Upsampling

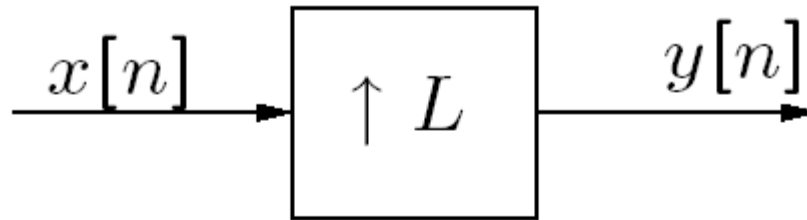


(b)



(c)

Upsampling

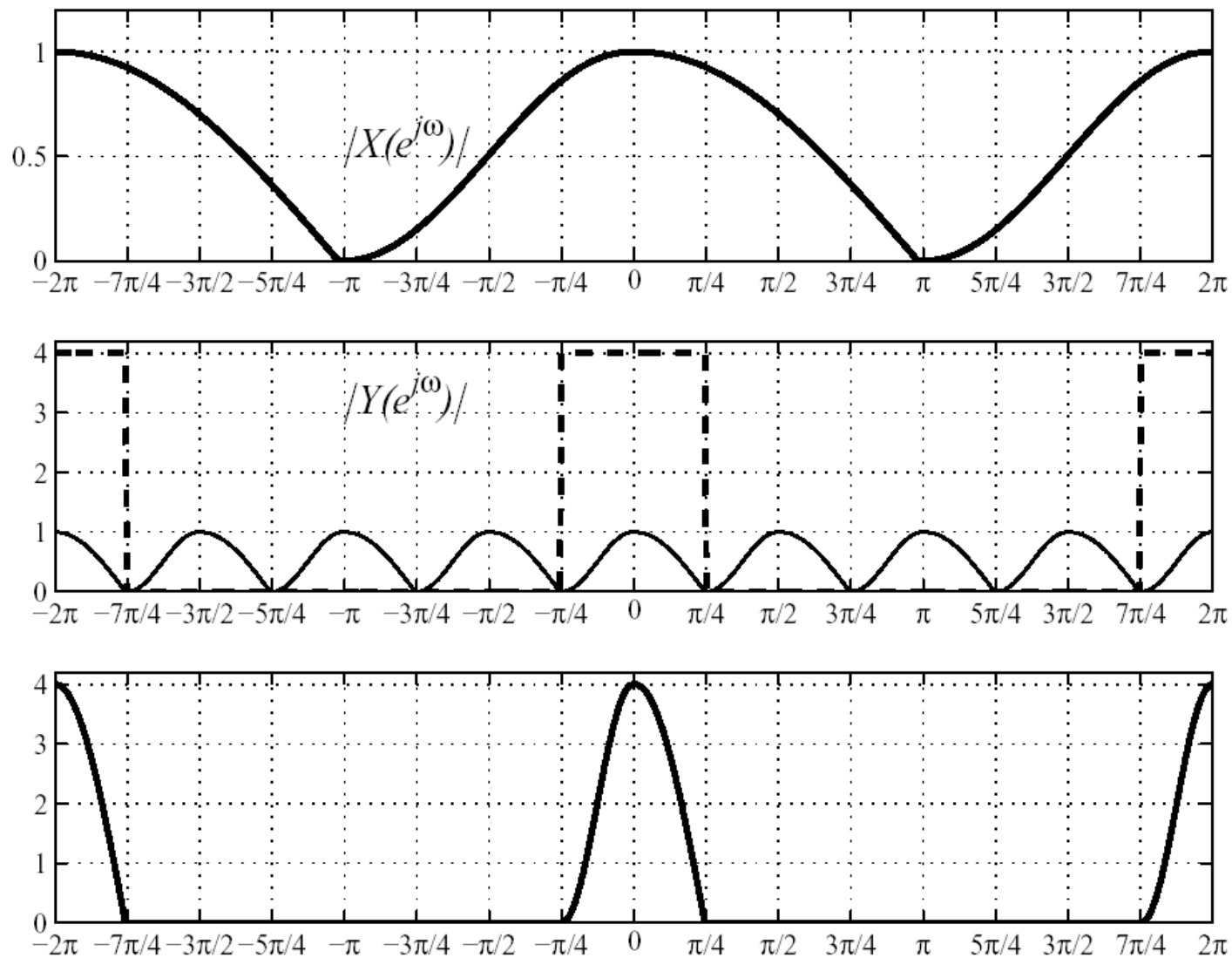


$$y[n] = \begin{cases} x[n/L] & n = kL \\ 0 & \text{otherwise} \end{cases}$$

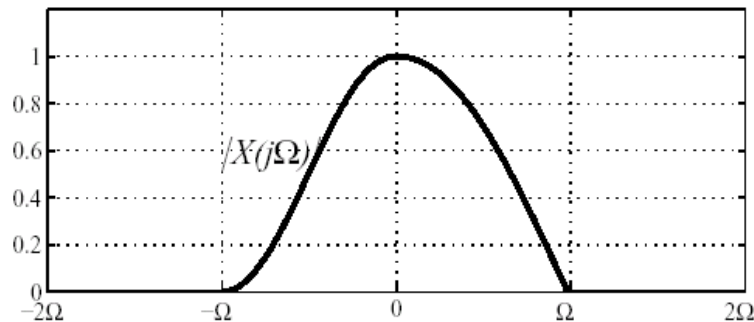
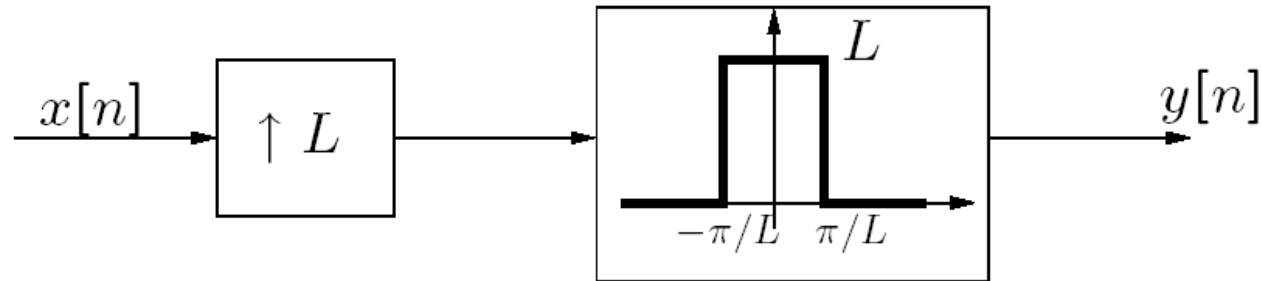
$$Y(e^{j\omega}) = X(e^{j\omega L})$$

$$Y(z) = X(z^L)$$

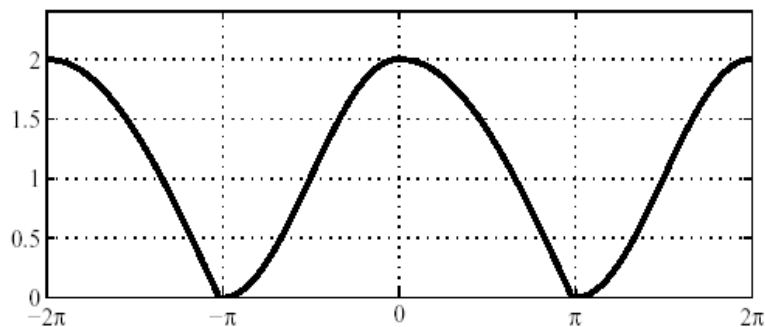
Upsampling in Freq Domain



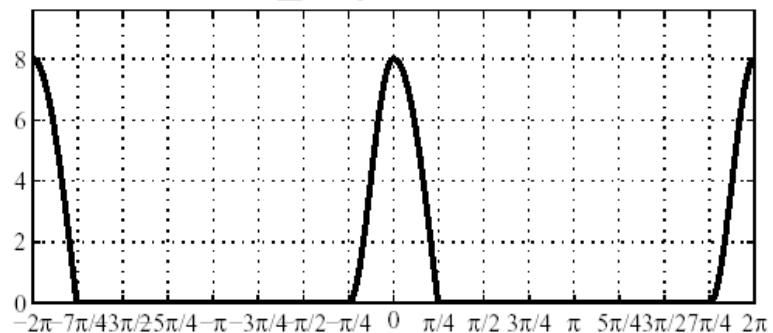
Upsampling in Freq Domain



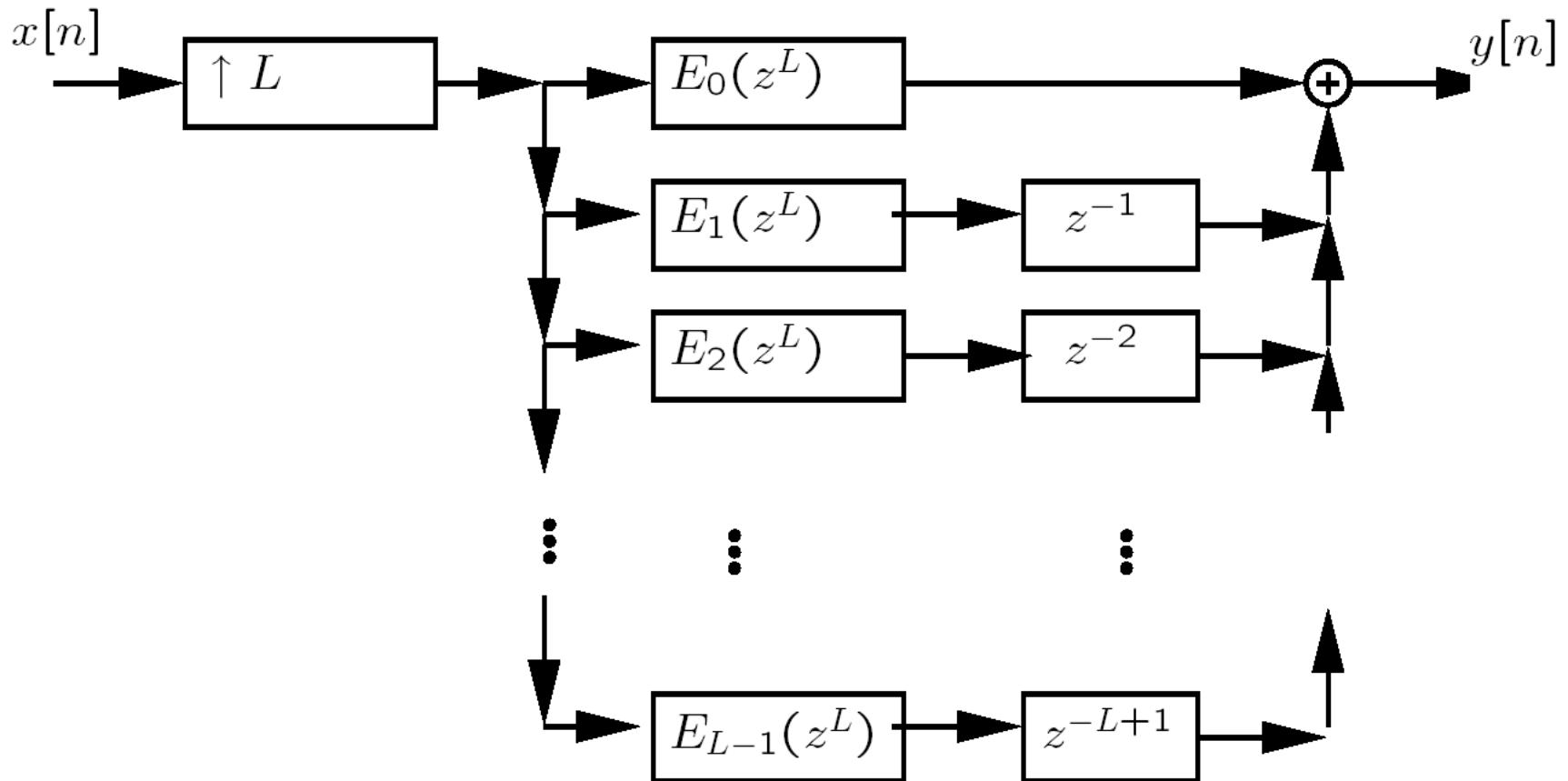
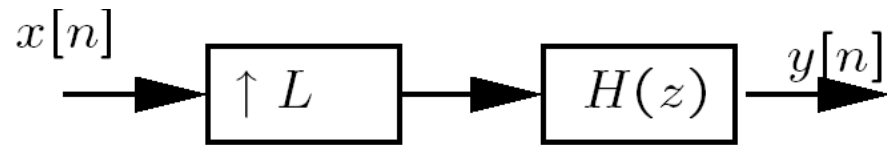
$$\frac{2\pi}{T} = 2\Omega_N$$



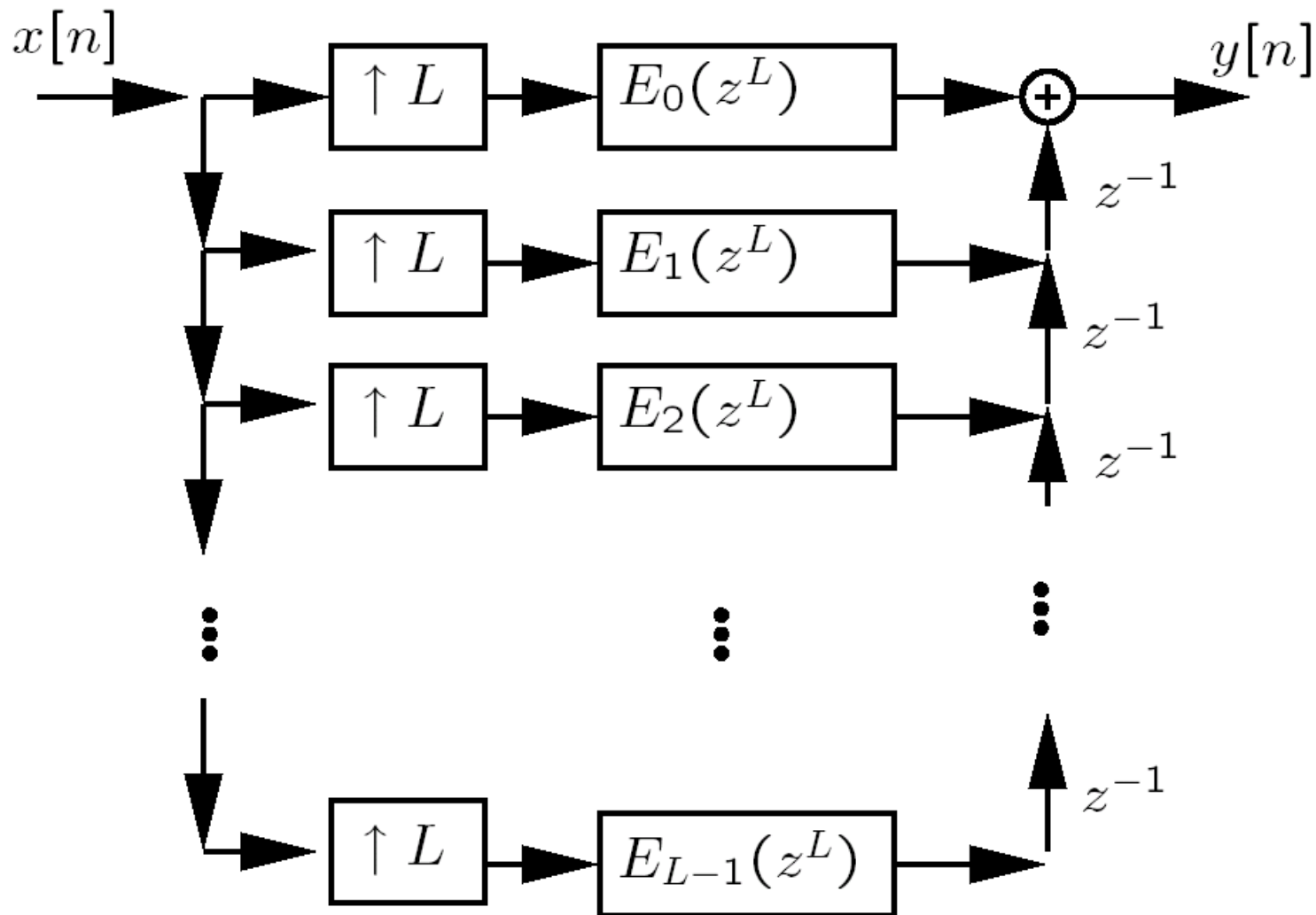
$$T' = \frac{T}{L}, \frac{2\pi}{T'} = 2L\Omega_N$$



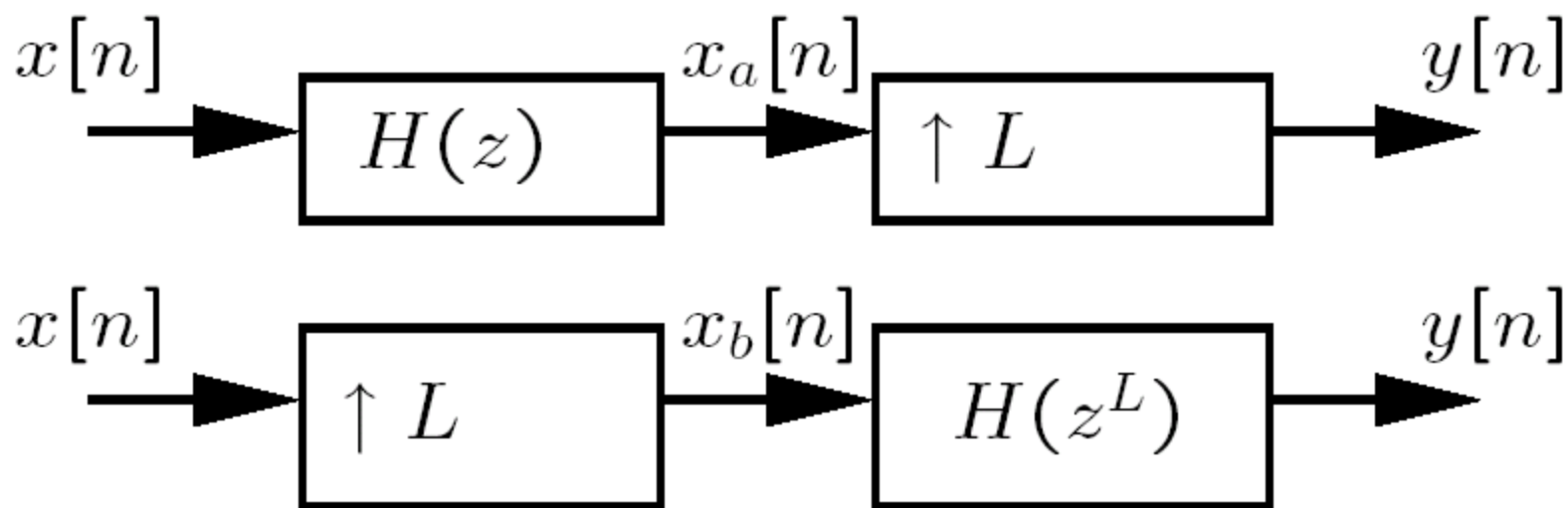
Polyphase Representation: Interpolation Filter



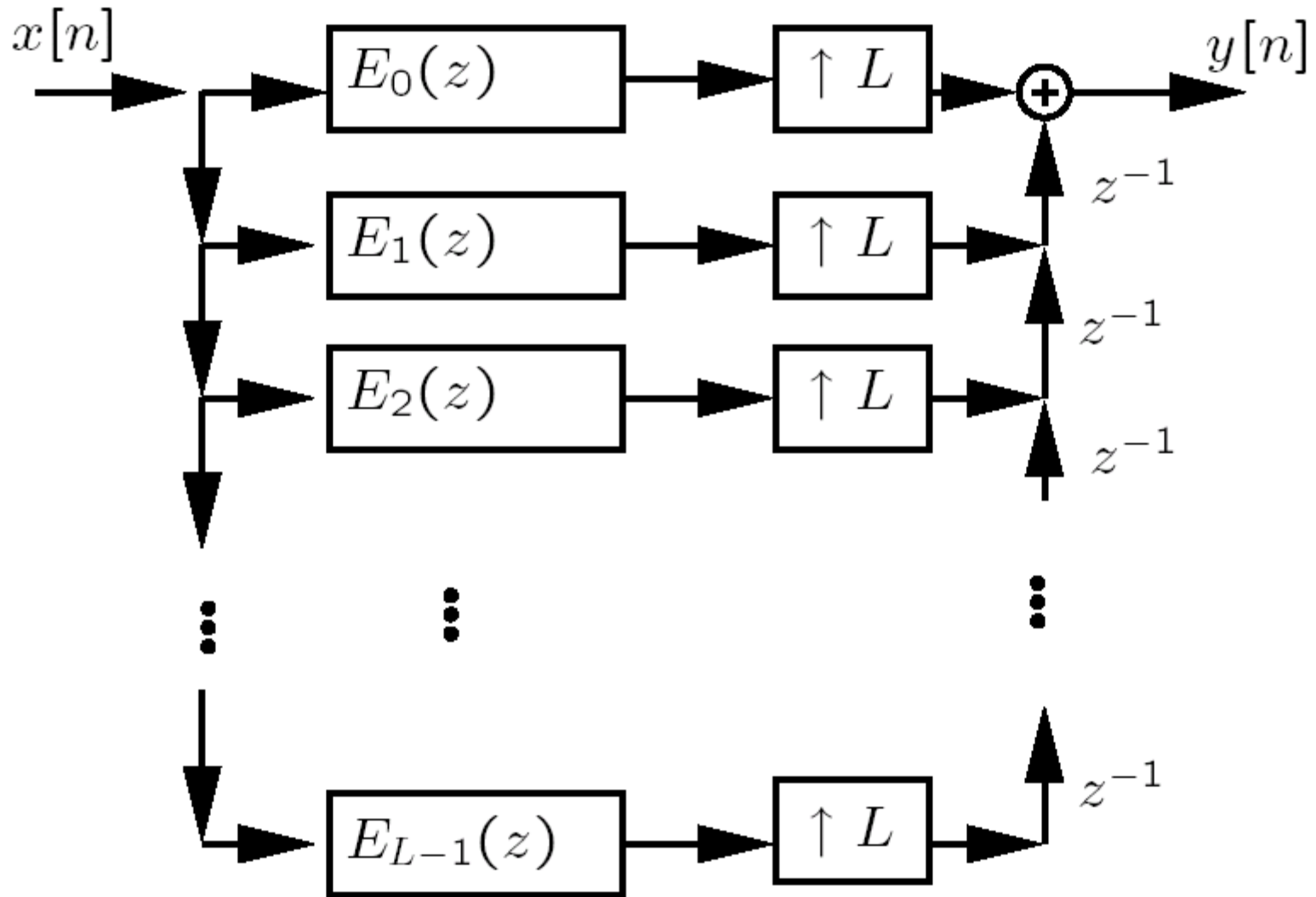
Polyphase Representation



Noble Identities



Polyphase Decomposition



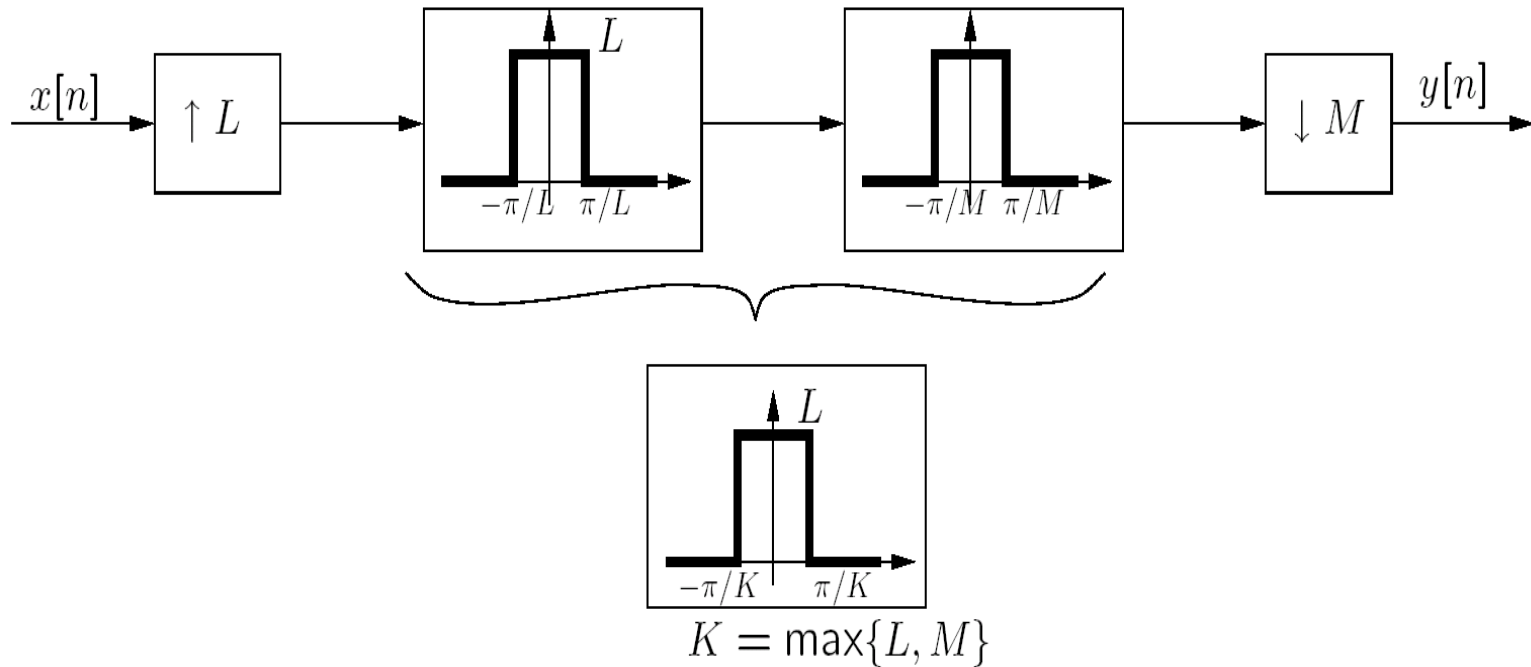
Interpolation Filter Implementation

- Ideal low-pass filter approximated
- Or replaced by linear interpolation:

$$h_{\text{lin}}[n] = \begin{cases} 1 - \frac{|n|}{L} & |n| \leq L \\ 0 & \text{elsewhere} \end{cases}$$

Sampling Rate Change by non-integer Factor

- Cascade up- and down-sampling:



Sampling Rate Change by non-integer Factor

$$L = 3, M = 2$$

